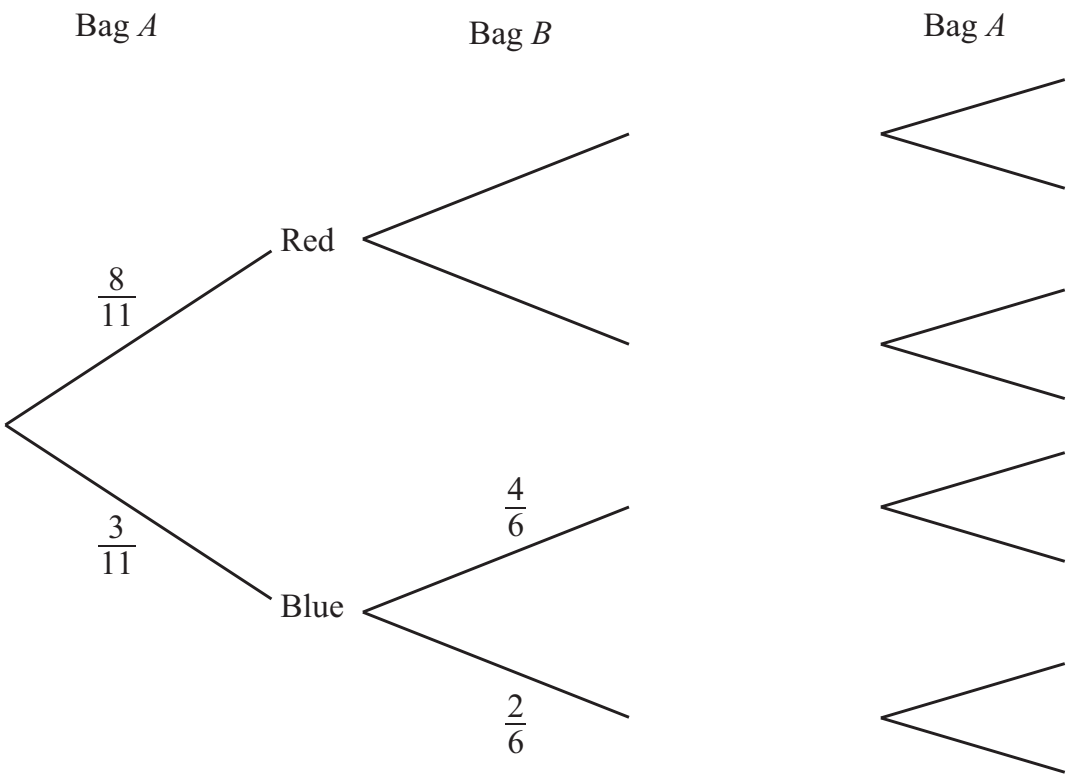
4 Bag *A* contains 8 red marbles and 3 blue marbles. Bag *B* contains 4 red marbles and 1 blue marble.

A marble is chosen at random from bag *A*. If the marble chosen is red, it is discarded. If the marble chosen is blue, it is placed in bag *B*.

A marble is then chosen at random from bag *B*. If the marble chosen is red, it is discarded. If the marble chosen is blue, it is placed in bag *A*.

A marble is now chosen at random from bag *A*.

(a) Complete the tree diagram below by entering all the remaining outcomes and probabilities. [3]



(b) Find the probability that all three marbles chosen are the same colour. [2]

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